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urlDict = 'https://raw.githubusercontent.com/cmuspinx/cmudict/master/
cmudict.dict';
urlFreq = 'https://raw.githubusercontent.com/sabuburi/Personal-Projects/refs/
heads/main/wordsearch%20(2).csv';
websave('cmudict.dict', urlDict);
websave('wordsearch.csv', urlFreq);

opts = detectImportOptions('wordsearch.csv');
opts.SelectedVariableNames = opts.VariableNames(1:2); % drops the empty
3rd column
freqVar = opts.VariableNames{2};
opts = setvartype(opts, freqVar, 'double');
opts = setvaropts(opts, freqVar, 'TreatAsMissing', 'NA');
opts.ImportErrorRule = 'omitrow'; % a header row is
repeated mid-file
us = readtable('wordsearch.csv', opts);
us.Properties.VariableNames(1:2) = {'Word', 'Log10Freq'};

us = rmmissing(us); % drop NA / bad rows
us.Word = lower(string(us.Word));
[~, i] = unique(us.Word, 'stable'); us = us(i,:); % drop duplicate
words
us.RawFreq = round(10.^us.Log10Freq - 1);

%% 3. CMU dictionary: parse and remove stress
L = readlines('cmudict.dict', 'Encoding', 'latin1');
L = strtrim(extractBefore(L + "#", "#")); % strip trailing
comments
L(L == "") = [];
word = lower(regexprep(extractBefore(L, " "), '\(\d+\)$', '')); % read(2)
-> read
pron = regexprep(extractAfter(L, " "), '\d', ''); %
AH0/AH1/AH2 -> AH
[~, i] = unique(word, 'stable'); % keep first
pronunciation
cmu = table(word(i), pron(i), 'VariableNames', {'Word', 'Pron'});

%% 4. Combine and discard unmatched words
T = innerjoin(us, cmu, 'Keys', 'Word');
fprintf('%d of %d frequency words matched\n', height(T), height(us));

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42888 of 57215 frequency words matched

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%% 5. Count each phoneme per word
vowels =
["AA", "AE", "AH", "AO", "AW", "AY", "EH", "ER", "EY", "IH", "IY", "OW", "OY", "UH", "UW"];
cons =
["B", "CH", "D", "DH", "F", "G", "HH", "JH", "K", "L", "M", "N", "NG", "P", "R", "S", "SH", "T",
"TH", "V", "W", "Y", "Z", "ZH"];

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phones = [vowels cons];

counts = zeros(height(T), numel(phones));
for r = 1:height(T)
    tok = split(T.Pron(r));
    [tf, loc] = ismember(tok, phones);
    assert(all(tf), 'Unknown phoneme in word "%s"', T.Word(r));
    counts(r,:) = accumarray(loc, 1, [numel(phones) 1]);
end

R = [T(:, {'Word', 'Log10Freq', 'RawFreq'}), array2table(counts,
'VariableNames', cellstr(phones))];
R.NumVowels      = sum(counts(:, 1:numel(vowels)), 2);
R.NumConsonants  = sum(counts(:, numel(vowels)+1:end), 2);
R.NumPhonemes    = R.NumVowels + R.NumConsonants;

%% 6. Summary: one row per phoneme
Type = [repmat("vowel", numel(vowels), 1); repmat("consonant", numel(cons),
1)];
S = table(phones', Type, sum(counts > 0)', sum(counts)', counts' *
R.RawFreq, ...
'VariableNames',
{'Phoneme', 'Type', 'WordsContaining', 'TotalOccurrences', 'FreqWeighted'});
S = sortrows(S, 'FreqWeighted', 'descend');

%% 7. Save
writetable(R, 'word_phoneme_counts.csv');
writetable(S, 'phoneme_summary.csv');
head(R), S

```

Word	Log10Freq	RawFreq	AA	AE	AH	AO	AW	AY	EH	ER	EY	IH
"a"	6.0175	1.0412e+06	0	0	1	0	0	0	0	0	0	0
"aardvark"	1.3424	21	2	0	0	0	0	0	0	0	0	0
"aback"	1.2041	15	0	1	1	0	0	0	0	0	0	0
"abacus"	1.1139	12	0	1	2	0	0	0	0	0	0	0
"abalone"	1.4314	26	0	1	1	0	0	0	0	0	0	0
"abandon"	2.617	413	0	1	2	0	0	0	0	0	0	0
"abandoned"	2.8319	678	0	1	2	0	0	0	0	0	0	0
"abandoning"	1.9395	86	0	1	2	0	0	0	0	0	0	1

S = 39×5 table

	Phoneme	Type	WordsContaining	TotalOccurrences	FreqWeighted
1	"T"	"consonant"	17330	20018	11863170
2	"AH"	"vowel"	20376	26600	11385839
3	"N"	"consonant"	15922	18738	9126553
4	"IH"	"vowel"	16391	19967	8098315
5	"S"	"consonant"	15944	18514	7085928
6	"IY"	"vowel"	9089	9702	6324962

	Phoneme	Type	WordsContaining	TotalOccurrences	FreqWeighted
7	"R"	"consonant"	13858	14877	6110444
8	"D"	"consonant"	11188	12402	5772049
9	"EH"	"vowel"	7737	8057	5459869
10	"L"	"consonant"	14574	15468	5256699
11	"M"	"consonant"	8294	8808	4778210
12	"UW"	"vowel"	3064	3089	4740521
13	"K"	"consonant"	12289	13492	4688036
14	"AY"	"vowel"	4165	4273	4509548
15	"DH"	"consonant"	255	255	3884111
16	"AE"	"vowel"	6714	6955	3839801
17	"W"	"consonant"	2458	2491	3808930
18	"AA"	"vowel"	5329	5506	3449762
19	"Y"	"consonant"	1670	1692	3172375
20	"Z"	"consonant"	8914	9318	2847480
21	"HH"	"consonant"	2043	2072	2846956
22	"ER"	"vowel"	8987	9397	2768552
23	"EY"	"vowel"	5761	5855	2707852
24	"B"	"consonant"	5876	6116	2680353
25	"OW"	"vowel"	3892	4037	2620363
26	"P"	"consonant"	8417	8910	2379427
27	"AO"	"vowel"	3255	3315	2356352
28	"F"	"consonant"	4823	4938	2324990
29	"V"	"consonant"	3553	3640	2169505
30	"G"	"consonant"	3365	3429	2019763
31	"NG"	"consonant"	4929	5036	1923939
32	"AW"	"vowel"	1182	1199	1191476
33	"SH"	"consonant"	3590	3626	886086
34	"TH"	"consonant"	922	928	847030
35	"UH"	"vowel"	726	732	799855
36	"JH"	"consonant"	2277	2314	725078
37	"CH"	"consonant"	1567	1579	537745
38	"OY"	"vowel"	409	409	125194
39	"ZH"	"consonant"	254	254	38787